

Predicting London House Prices Using Machine Learning

Capstone Executive Summary and Report

https://github.com/MKA90/Capstone_Project_London_House_Price_Prediction/tree/main

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Executive Summary

This project developed a machine learning framework for predicting London residential property prices using structural, geographic, temporal and historical housing data. Accurate house price prediction is valuable across a wide range of real estate and financial applications because it enables more informed valuation decisions, improves mortgage risk assessment, supports property investment analysis, enhances housing market forecasting, and provides deeper insight into the geographic and economic factors influencing residential property markets.

Using over 418,000 London property transaction records spanning from 1995 to 2024, the project conducted extensive exploratory data analysis (EDA), feature engineering, ensemble modelling, inflation-adjusted pricing models and segmented market modelling to predict historical residential property prices while improving predictive accuracy and maintaining interpretability and economic meaning.

The project progressed through multiple modelling stages including baseline regression models, Decision Trees, K-Nearest Neighbours, Random Forest, XGBoost, LightGBM, inflation-adjusted modelling frameworks, and segmented market models. Feature engineering played an important role in both improving dataset representation and enhancing predictive performance. New variables including total rooms, bathrooms per bedroom, square metres per room, distance to Central London, latitude-longitude interaction terms, sale year, and years since sale were introduced to better capture structural, temporal, and spatial housing relationships within the dataset. Geographic information was also enhanced using target encoding of London outcodes, which initially improved predictive performance for several nonlinear models during the earlier modelling stages.

Initial global models substantially improved upon the baseline Dummy Regressor, which produced a validation Root Mean Squared Error (RMSE) of approximately £791,000. However, the strongest performance was achieved through segmented market modelling, where the London housing market was divided into low-, medium-, and high-priced market clusters before training separate models within each segment. The best-performing segmented Random Forest models achieved test RMSE values of approximately £80,000 within the lower-priced housing cluster, approximately £431,000 within the medium-priced cluster, and approximately £529,000 within the higher-priced housing cluster, substantially outperforming earlier global modelling approaches.

A major contribution of the project was the investigation of inflation adjusted property prices. Earlier nominal-price models relied heavily on temporal variables such as sale year and years since sale, indicating that historical market appreciation contributed strongly to predictive performance. After adjusting prices using CPI inflation data and reducing temporal market effects, predictive performance declined considerably, demonstrating that a substantial proportion of predictive accuracy within nominal price models originated from long term market appreciation rather than purely intrinsic property characteristics. Although inflation adjusted models provided useful economic insight, nominal price models incorporating temporal variables ultimately produced substantially stronger predictive performance and were therefore retained within the final modelling framework.

SHAP (SHapley Additive exPlanations) analysis confirmed that the final cluster specific models learned economically meaningful pricing relationships. Across the low-, medium-, and high-priced market segments, floor area, geographic location (latitude and longitude), temporal variables and proximity to Central London emerged as the most influential predictors of property prices. However, the relative importance of these drivers differed substantially across market segments, indicating that housing submarkets respond to different pricing mechanisms.

Overall, the project delivered a transparent, reproducible and commercially relevant machine learning framework for London house price prediction while also providing strong economic interpretation, critical evaluation of modelling choices and insight into the challenges associated with modelling highly heterogeneous real-estate markets.

Introduction

Accurately predicting residential property prices is important for estate agencies, mortgage lenders, institutional investors and individual buyers because it supports pricing decisions, investment analysis and market forecasting. London presents a particularly challenging environment for valuation due to substantial variation in geography, property characteristics, and long-term market appreciation.

The dataset used in this project contains approximately 418,000 London residential property records spanning from 1995 to 2024. The dataset includes structural property characteristics such as floor area, bedrooms, bathrooms, tenure, property type, energy ratings and geographic coordinates alongside historical transaction prices.

This project aimed to develop a transparent and reproducible machine learning pipeline capable of predicting London residential property prices using structural, geographic, and historical transaction data. The analysis incorporated exploratory data analysis, feature engineering, machine learning experimentation, inflation-adjusted modelling, and segmented market modelling.

Methodology

The dataset used in this project consists of approximately 418,000 London residential property transactions spanning from 1995 to 2024. The data was obtained from the Kaggle dataset [“London House Price Data”](#) by Jake Wright, which is likely derived from HM Land Registry Price Paid Data enriched with additional publicly available housing information.

The dataset contains structural property characteristics, geographic information, historical transaction records, and energy efficiency ratings. Key variables include floor area, number of bedrooms, bathrooms, living rooms, property type, tenure, latitude, longitude, energy ratings, historical sale date, and historical sale price.

Several pricing-related variables originally included in the dataset, such as rental estimates and historical percentage price changes, were removed during preprocessing due to substantial risks of data leakage. These variables were considered inappropriate for realistic predictive modelling because they contained information closely related to the target variable.

	fullAddress	postcode	country	outcode	latitude	...	saleEstimate_valueChange.saleDate	history_date	history_price	history_percentageChange
0	Flat 9, 35 Furnival Street, London, EC4A 1JQ	EC4A 1JQ	England	EC4A	51.517282	...	2010-03-30	1995-01-02	830000	NaN
1	Flat 6, 35 Furnival Street, London, EC4A 1JQ	EC4A 1JQ	England	EC4A	51.517282	...	2000-05-26	1995-01-02	830000	NaN
2	Flat 35, Octavia House, Medway Street, London,...	SW1P 2TA	England	SW1P	51.495505	...	2019-09-04	1995-01-03	249950	NaN
3	24 Chester Court, Lomond Grove, London, SE5 7HS	SE5 7HS	England	SE5	51.478185	...	2024-01-25	1995-01-03	32000	NaN
4	18 Alexandra Gardens, London, N10 3RL	N10 3RL	England	N10	51.588774	...	2022-12-14	1995-01-03	133000	NaN

5 rows × 28 columns

Figure 1: Preview of the London housing dataset.

Target Variable and Evaluation Metrics

The target variable for prediction was historical residential property price (`history_price`). Two overarching modelling approaches were explored throughout the project:

Raw-price modelling using nominal historical transaction prices

Inflation-adjusted modelling using CPI-adjusted prices

Model performance was evaluated primarily using Root Mean Squared Error (RMSE), which measures average prediction error while penalising larger pricing mistakes more heavily. Additional evaluation metrics included validation R^2 scores, Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and overfitting gaps where appropriate.

A Dummy Regressor predicting the mean property price was used as the baseline benchmark for the initial raw-price modelling approach, producing a validation RMSE of approximately £791,000.

Preprocessing & Exploratory Data Analysis

The raw dataset contained a mixture of numerical, categorical, geographic and historical transaction variables. Initial exploratory data analysis (EDA) was conducted to investigate missingness patterns, target variable distributions, feature distributions, outliers, correlations and geographic pricing trends within the London housing market. Following EDA and feature engineering, the dataset was split into training, validation, and test sets before preprocessing steps were applied to reduce the risk of data leakage.

A small number of duplicate observations were identified and removed as they provided no additional information and could artificially inflate predictive performance.

Feature Engineering

Several engineered variables were introduced to better capture structural, spatial, and temporal relationships not fully represented by the original dataset. Temporal variables such as `sale_year` and `sale_month` were created to capture long-term market trends and seasonal housing patterns. Structural variables including `total_rooms`, `bathrooms_per_bedroom`, and `sqm_per_room` provided more informative measures of property size and housing quality.

Additional spatial variables included `lat_lon_interaction` and `distance_to_central_london_km`. The latter was calculated using the Haversine formula to measure proximity to Central London. These engineered variables aimed to provide richer and more interpretable predictive signals for later modelling stages.

Missing Value Analysis

Significant missingness was identified within variables such as `bedrooms`, `livingRooms`, `floorAreaSqM`, and `currentEnergyRating`. Missingness analysis suggested that missing values were concentrated primarily within structural property characteristics rather than being randomly distributed across the dataset.

Engineered variables such as `bathrooms_per_bedroom`, `total_rooms`, and `sqm_per_room` also inherited missingness from their source variables. Although several variables contained relatively high proportions of missing data, these features represented important structural characteristics of the properties and were therefore retained for further experimentation.

Several preprocessing approaches were evaluated to determine the most appropriate method for handling missing data. These included complete case analysis, median imputation for numerical variables, most-frequent categorical imputation, explicit missing category assignment, and KNN imputation. However, all imputation methods reduced predictive performance relative to complete case analysis. This likely occurred because missing structural property characteristics could not be reliably inferred within London’s highly heterogeneous housing market. As a result, the final modelling pipeline adopted complete case analysis.

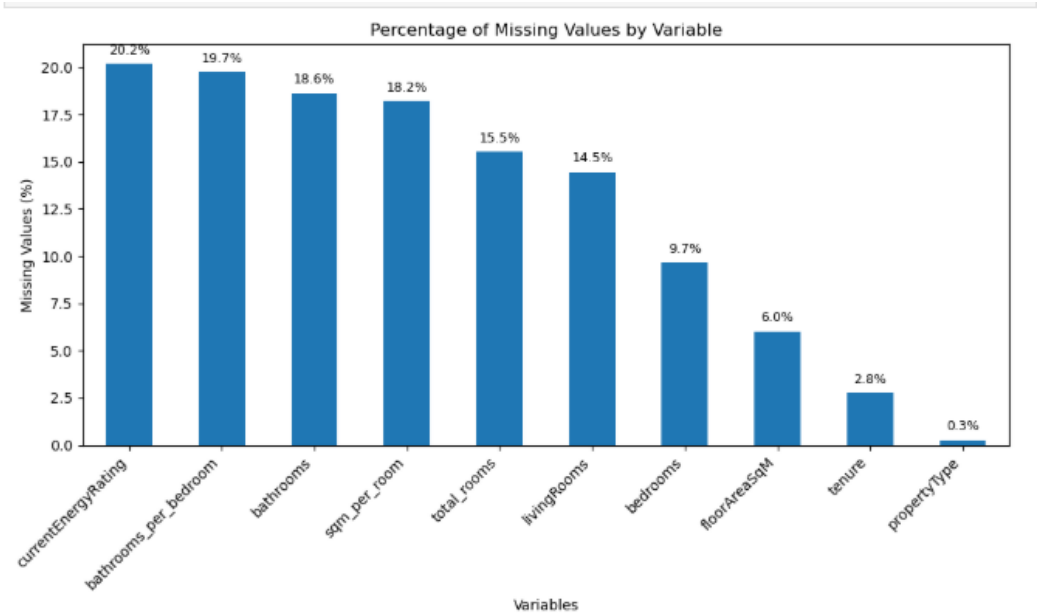


Figure 2: Percentage of missing values by feature.

Target Distribution Analysis

The distribution of the target variable, `history_price`, was examined to better understand its scale, spread, and potential skewness. Initial analysis showed that house prices were heavily positively skewed with a long right tail caused by extremely high value London properties. Boxplot analysis also confirmed the presence of substantial luxury-property outliers.

To reduce the influence of extreme observations, a log transformation of `history_price` was investigated. The transformed distribution produced a substantially more symmetric shape and improved target stability, which was expected to benefit several modelling approaches, particularly linear models.

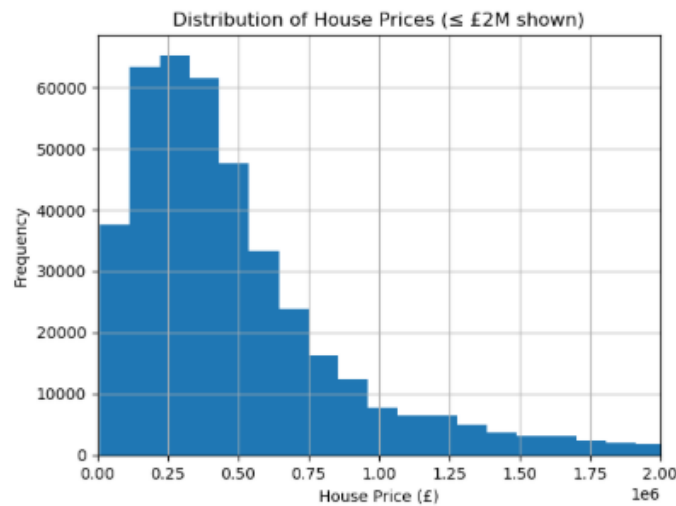


Figure 3: Distribution of London house prices, showing positive skew.

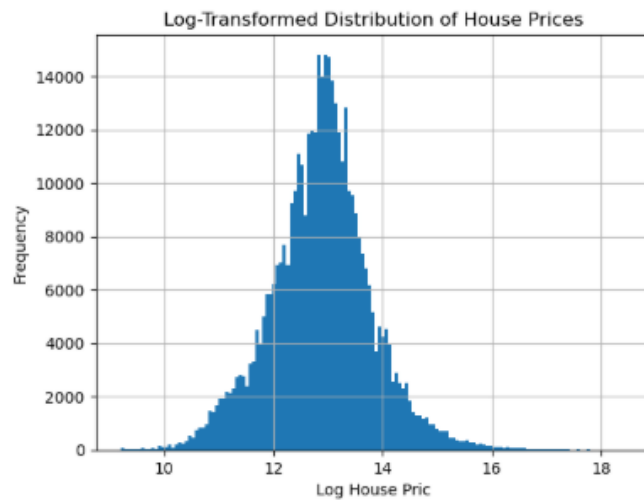


Figure 4: Log-transformed distribution of London house prices.

Numerical and Categorical Variable Analysis

Several numerical variables, particularly `floorAreaSqM`, `livingRooms`, `bathrooms`, and `total_rooms`, exhibited substantial positive skewness, reflecting the presence of a relatively small number of exceptionally large or high value London properties. This reinforced earlier observations regarding extreme property values and supported the use of log transformations and robust nonlinear modelling approaches.

Count based variables such as bedrooms and bathrooms displayed discrete distributions characteristic of residential housing data, suggesting that structural housing features may influence prices in nonlinear ways rather than through simple linear relationships.

Categorical analysis revealed noticeable class imbalance across several variables. Within propertyType, a relatively small number of housing categories accounted for the majority of observations, while categories such as bungalows and detached houses were comparatively rare. Similar imbalance patterns were observed within tenure and currentEnergyRating. The outcode variable also exhibited high cardinality, highlighting the substantial geographic diversity of the London housing market. These findings later informed the selection of encoding strategies and supported the use of flexible tree-based ensemble models capable of handling heterogeneous housing characteristics and complex spatial relationships.

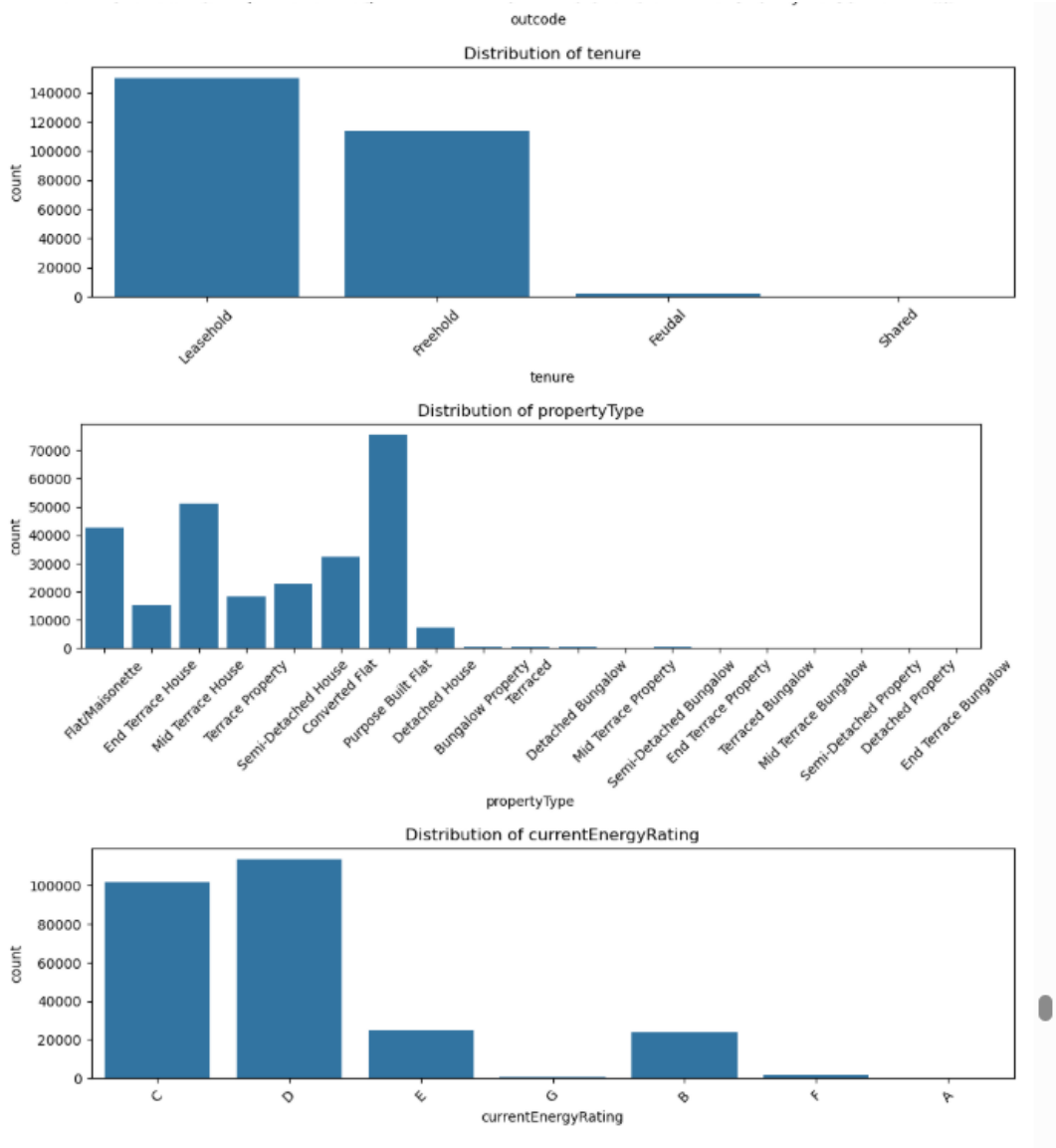


Figure 5: Distribution of key categorical variables.

Correlation and Geographic Analysis

Correlation analysis identified moderate positive relationships between structural housing characteristics and property prices. Variables associated with property size, including floorAreaSqM, total_rooms, bedrooms, and bathrooms, all demonstrated positive correlations with history_price, confirming that larger properties generally achieved higher valuations.

Geographic analysis also highlighted the importance of spatial location within the London housing market. distance_to_central_london_km exhibited a negative relationship with property prices, suggesting that properties located further from Central London tended to have lower valuations. Longitude also demonstrated moderate relationships with both property prices and distance from Central London, reflecting broader spatial pricing patterns across London boroughs.

Several structural variables displayed relatively strong intercorrelations, particularly between bedrooms, total_rooms, and floorAreaSqM, indicating overlapping measures of property size and composition. These findings supported the later use of flexible tree-based ensemble models capable of handling correlated predictors and nonlinear pricing relationships.

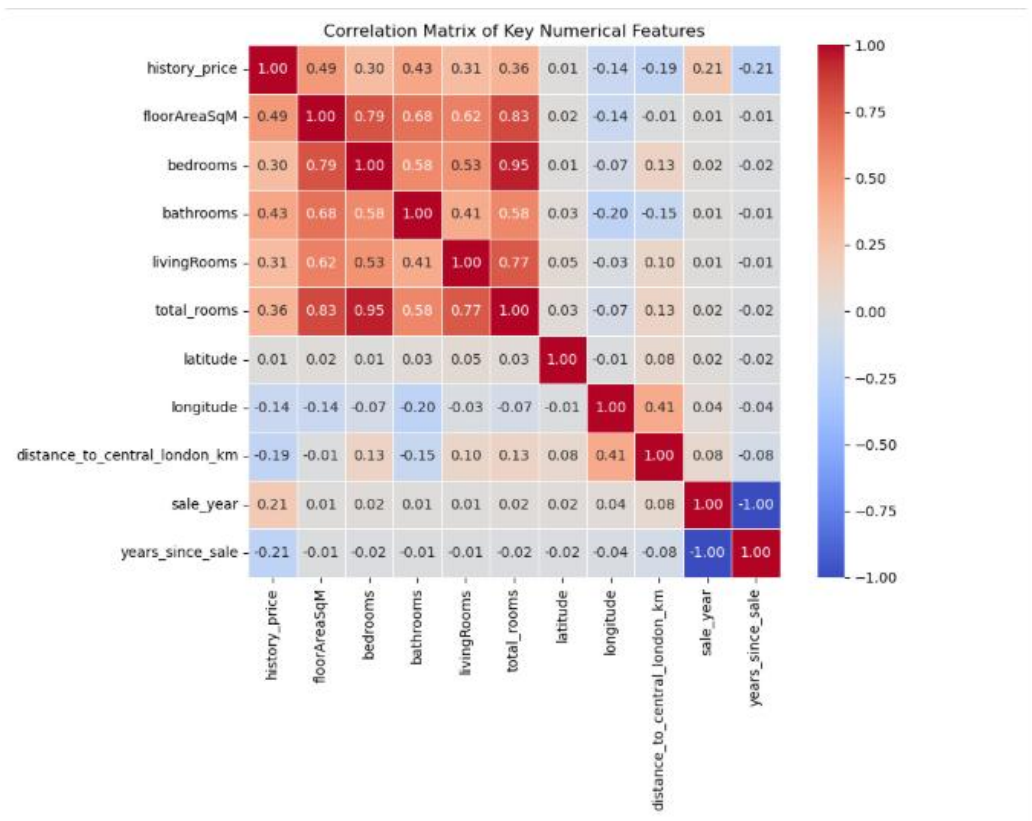


Figure 6: Correlation matrix of key numerical features.

Geographic analysis showed substantial spatial variation in property prices across London. Properties located closer to Central London generally achieved higher valuations, while prices tended to decline as distance from the city centre increased. This geographic pricing gradient supported the inclusion of spatial variables such as latitude, longitude, outcode encoding, and distance_to_central_london_km within later modelling stages.



Figure 7: House prices versus distance to Central London.

Modelling Approach

The modelling process progressed from simple baseline regression models to more advanced ensemble and segmented market frameworks to improve predictive accuracy across the London housing market.

Initial experiments included baseline and linear approaches such as Dummy Regression, Linear Regression, Ridge Regression and Lasso Regression before progressing to nonlinear methods including Decision Trees, K-Nearest Neighbours (KNN), Random Forest, XGBoost, and LightGBM.

Additional preprocessing and feature engineering techniques were introduced throughout the project, including target encoding of London outcodes, log-transformed target variables, outlier trimming, and engineered spatial, structural and temporal variables such as total rooms, distance to Central London and years since sale.

The project also explored inflation adjusted property price modelling using CPI-adjusted prices to investigate the influence of long-term market appreciation on predictive performance. Finally, segmented market modelling approaches were introduced by dividing the housing market into low-, medium-, and high-priced clusters before training separate models within each segment.

Model Findings

A Dummy Regressor predicting the mean historical property price established the baseline benchmark with a validation RMSE of approximately £791,000. Linear and regularised regression models improved upon this baseline but generally underperformed relative to later nonlinear and ensemble approaches.

Decision Tree and KNN models produced some of the strongest early predictive performance by capturing nonlinear housing relationships and local property similarities, although standalone Decision Trees demonstrated substantial overfitting. Target encoding of London outcodes improved performance for the initial nonlinear models by helping capture local geographic pricing patterns.

Early experiments showed that log transformation and selective outlier trimming could substantially improve predictive performance by reducing target skewness and limiting the influence of extreme luxury property observations. However, the final modelling framework prioritised retaining the full housing market distribution

to preserve commercial realism and generalisability. Ensemble models including Random Forest, XGBoost, and LightGBM ultimately achieved the strongest overall predictive performance by capturing complex structural, geographic, and temporal relationships within the housing data.

Inflation adjusted modelling demonstrated that a substantial proportion of predictive accuracy within nominal-price models originated from long-term market appreciation trends rather than purely intrinsic property characteristics. However, because the nominal-price models consistently produced stronger predictive performance and more commercially useful valuation estimates, the final modelling framework reverted to using historical nominal sale prices (`history_price`) as the target variable for the final model selection process.

Results

Final Model Selection

The strongest performing models were achieved using segmented market modelling combined with tuned Random Forest frameworks. Dividing the London housing market into low-, medium-, and high-priced clusters substantially improved predictive performance by reducing market heterogeneity and allowing models to specialise within narrower pricing ranges.

Market Segment	Final Model	Validation RMSE	Test RMSE	Validation R ²	Test R ²
Low-price cluster	Random Forest	£64,690	£79,753	0.859	0.797
Medium-price cluster	Random Forest	£318,975	£431,000	0.650	0.457
High-price cluster	Random Forest	£432,766	£529,308	0.871	0.789

Table 1: Final segmented Random Forest model performance across the London housing market clusters

The final low-price cluster Random Forest model achieved a test RMSE of approximately £79,753 with a test R² of 0.80, while the medium- and high-price cluster models achieved test RMSE values of approximately £431,000 and £529,000 respectively. Although predictive accuracy declined within higher-priced market segments, this likely reflects the substantially greater heterogeneity, volatility, and uniqueness present within London’s luxury housing market.

Overall, the findings demonstrated that combining feature engineering, geographic information, ensemble modelling and segmented market frameworks produced the most accurate and commercially relevant predictive models for historical London property price estimation.

Model Interpretability and SHAP Analysis

SHAP (Shapley Additive Explanations) analysis was used to improve model interpretability and identify the variables that contributed most strongly to predicted property prices within each segmented housing cluster. The SHAP results demonstrated that different market segments relied on different predictive drivers, further supporting the decision to use segmented modelling approaches rather than a single global model.

Within the low-price housing cluster, geographic and temporal variables including longitude, sale year, and years since sale were among the strongest drivers of predicted prices, alongside structural variables such as

bathrooms and floor area. This suggests that lower-priced London housing remained highly sensitive to both local geographic variation and broader market appreciation trends.

Table X: Top 5 SHAP Features – Low-Price Cluster

	Rank	Feature	Relative Importance (%)
0	1	Longitude	18.8400000000
1	2	Sale year	17.2000000000
2	3	Years since sale	15.2300000000
3	4	Bathrooms	14.9300000000
4	5	Latitude	13.7600000000

Table 2: Top five SHAP features for the low-price housing cluster.

The medium-price housing cluster demonstrated a stronger reliance on spatial and structural variables. Distance to Central London and floor area became substantially more influential within this segment, indicating that accessibility and property size played a larger role in determining medium-priced housing valuations. Geographic location variables remained highly important, although temporal variables became relatively less dominant compared with the lower-priced segment.

Table X: Top 5 SHAP Features – Low-Price Cluster

	Rank	Feature	Relative Importance (%)
0	1	Longitude	18.8400000000
1	2	Sale year	17.2000000000
2	3	Years since sale	15.2300000000
3	4	Bathrooms	14.9300000000
4	5	Latitude	13.7600000000

Table 3: Top five SHAP features for the medium-price housing cluster.

Within the high-price housing cluster, floor area emerged as the dominant predictive variable alongside proximity to Central London. Temporal variables including sale year and years since sale continued to contribute strongly, while structural room-count variables became comparatively less influential. These findings suggest that premium London housing markets are driven more heavily by location, scale, and long-term market appreciation than by simple room-based structural characteristics.

	Rank	Feature	Relative Importance (%)
0	1	Floor area	24.0100000000
1	2	Distance to Central London	20.9400000000
2	3	Sale year	15.4300000000
3	4	Years since sale	15.2500000000
4	5	Longitude	13.8900000000

Table 4: Top five SHAP features for the high-price housing cluster.

Overall, the SHAP analysis confirmed that the segmented models captured different housing market dynamics across pricing ranges and improved transparency by identifying the variables that most strongly influenced predicted property prices within each segment.

Conclusion

This project successfully developed a reproducible machine learning framework for predicting historical London residential property prices using structural, geographic, temporal, and historical transaction data. Across multiple modelling stages, nonlinear and ensemble approaches consistently outperformed simpler linear models, with segmented market modelling producing the strongest predictive performance overall.

The findings demonstrated that preprocessing and feature engineering played a major role in improving model accuracy. Techniques including log transformation, outlier trimming, engineered spatial variables, and geographic encoding substantially enhanced predictive performance by allowing models to better capture the structural and spatial complexity of the London housing market. In contrast, missing value imputation methods generally reduced predictive accuracy relative to complete case analysis, suggesting that missing structural housing information could not be reliably inferred within such a heterogeneous market.

The project also demonstrated the importance of temporal market trends within historical house price prediction. Inflation adjusted modelling showed that nominal price models relied heavily on long-term market appreciation effects rather than purely intrinsic property characteristics. However, retaining temporal variables within the final nominal price models ultimately produced stronger predictive performance and greater commercial relevance for practical valuation applications.

Segmented market modelling further highlighted the substantial heterogeneity present across London's housing market. Lower- and medium-priced housing segments proved considerably easier to predict than extreme luxury property markets, where large price variability and unique property characteristics continued to generate relatively high prediction error despite strong model performance overall.

SHAP analysis of the final segmented Random Forest models provided additional interpretability by identifying the most influential drivers of predicted prices within each market segment. Geographic location, floor area, distance to Central London, and temporal variables consistently emerged as dominant predictors, although the relative importance of features varied across low-, medium-, and high-priced clusters. This reinforced the value of segmented modelling, as different areas of the market were influenced by different structural and geographic factors.

Overall, the project demonstrated that combining feature engineering, geographic information, ensemble machine learning, and segmented market frameworks can produce commercially useful and interpretable predictive models for London residential property valuation. Future improvements could incorporate richer

neighbourhood and property condition information, including transport accessibility, renovation quality, school performance, crime rates, and local economic indicators, to further improve predictive performance and market interpretation.